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Dorsal-ventral dissociation of suprachiasmatic nucleus oscillators reveals a potential mechanism for emergent multidien rhythms

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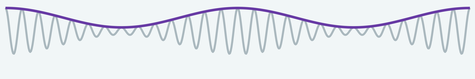
NSTC 國家科學及技術委員會 National Science and Technology Council

Circadian: about a day (~24 h)



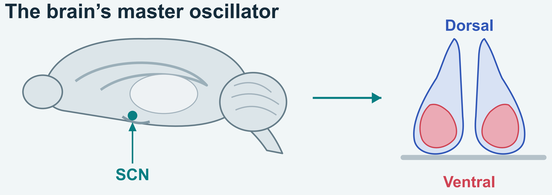
A circadian rhythm in activity.
It keeps going even in
constant darkness.

Infradian: days to weeks

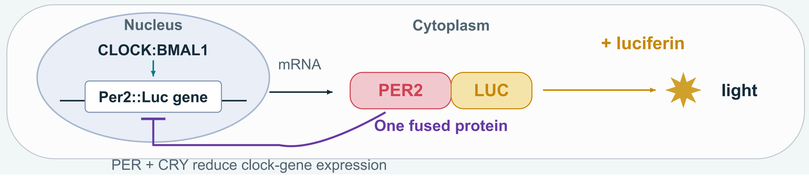


Circadian activity rises and falls
more slowly, across many days.
Could circadian oscillators make this?

The brain's master oscillator



Tracking PER2 protein rhythms



1 Behavior

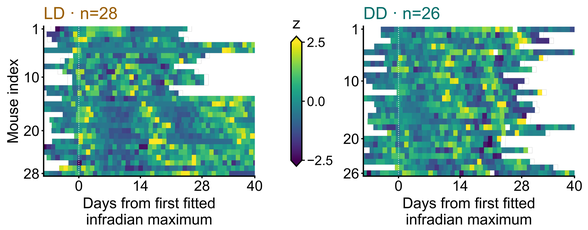
Multi-day activity rhythms show no consistent Moon or magnetic links.



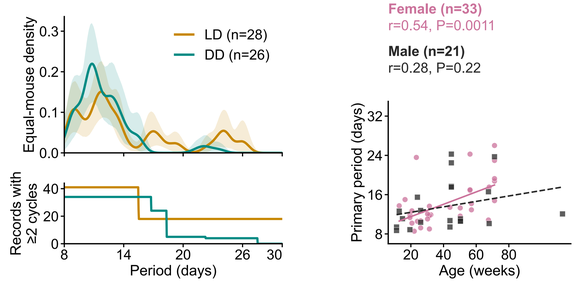
41 mice in LD
Wheel-running
12 h light / 12 h dark



34 mice in DD
Locomotor activity
Constant darkness

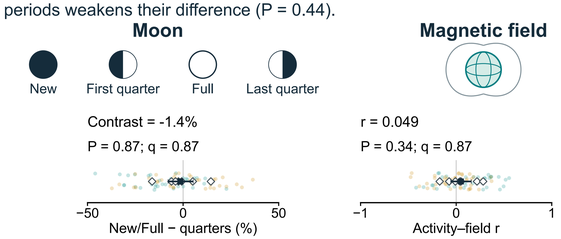


One row per mouse. Color shows circadian amplitude (within-mouse z-score). First fitted infradian maxima are aligned; time is not stretched. White: no observation.



54 / 75 mice had robust periodicity >8 days. Each contributing mouse has equal weight in the period distribution.

Longer rhythms need longer records. Restricting both groups to detectable periods weakens their difference (P = 0.44).

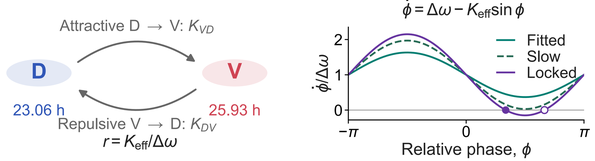


Moon: New/Full-minus-quarter activity (% of each mouse's mean). Magnetic field: daily activity–field correlation.

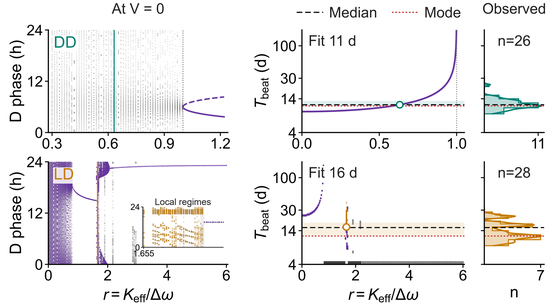
2 Theoretical mechanism

Coupled circadian oscillators can make a slow beat.

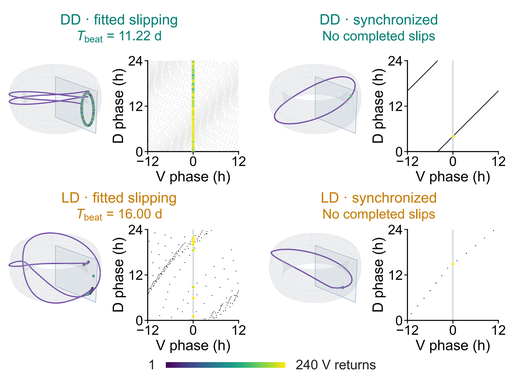
$$\dot{\theta}_D = \omega_D - K_{DV}\sin(\theta_V - \theta_D) \qquad \dot{\theta}_V = \omega_V + K_{VD}\sin(\theta_D - \theta_V)$$
$$\phi = \theta_D - \theta_V \qquad \Delta\omega = \omega_D - \omega_V \qquad K_{\text{eff}} = K_{VD} - K_{DV}$$
$$\dot{\phi} = \Delta\omega - K_{\text{eff}}\sin\phi$$



Coupling shifts when each oscillator peaks. If their phase difference keeps changing, a slower beat appears.



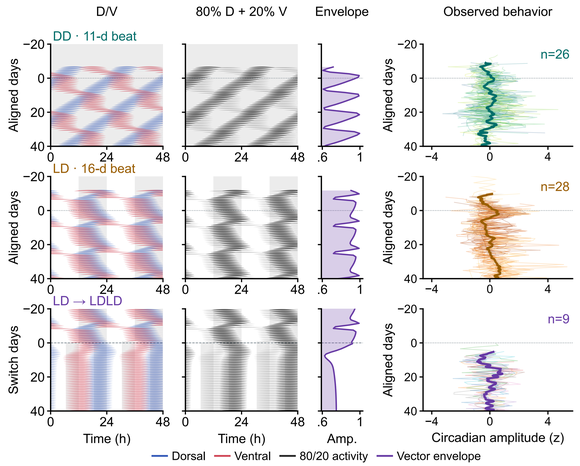
Beating slows near phase locking. On the torus below, a slipping trajectory keeps moving around; a locked pair returns to the same phase.



Dots mark returns to V phase 0. Locked pairs stop slipping.

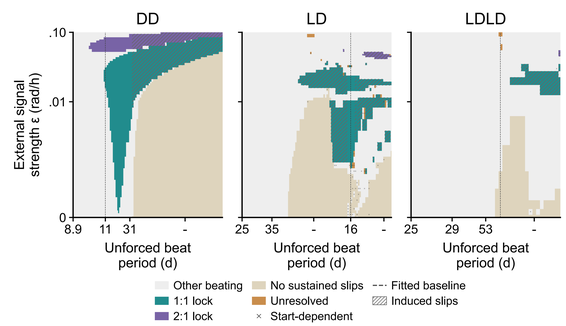
3 Model and behavior

Light schedules change beating and the D–V phase difference.



Black: simulated 80% D + 20% V activity. Thin traces: mice; bold: mean ± SEM. LDLD dips are aligned for display only.

Can beating be synchronized by a slow external signal?



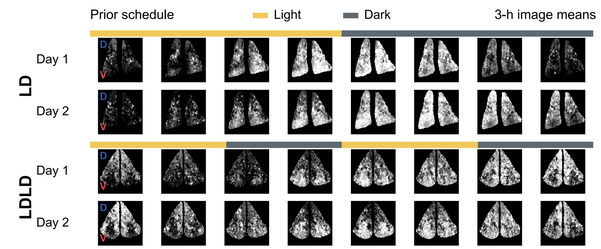
Green/purple: 14.77-day synchronization. Period ticks keep coupling spacing; a dash means no unforced beat.

A slow signal can synchronize beating in the model. The response depends on the light schedule.

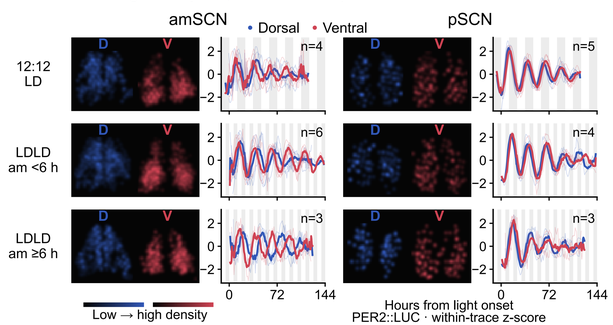
4 SCN oscillators

More separated anterior oscillators accompany clearer activity bouts.

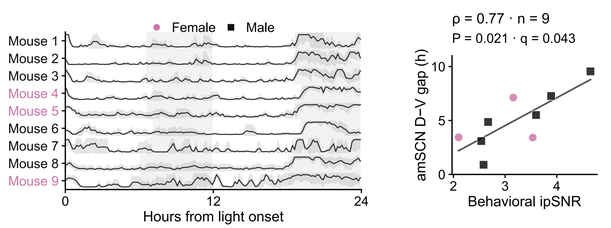
LDLD: two light/dark cycles in each 24 h.



PER2::LUC through two cycles; bars show the prior schedule.



Dissected SCN slices keep oscillating outside the animal. Gray: prior darkness. 3/9 LDLD anterior samples had D–V gaps ≥6 h.



Seven pre-collection days (mean ± SD). ipSNR compares peaks with inter-peak variability; its SCN association is exploratory.

Take-home message

Among mice with robust rhythms, median periods were 11 d in DD and 16 d in LD. In the model, circadian oscillators that slip past each other generate slow rhythms without an external multi-day signal.

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